



AI VIET NAM

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Ensembling

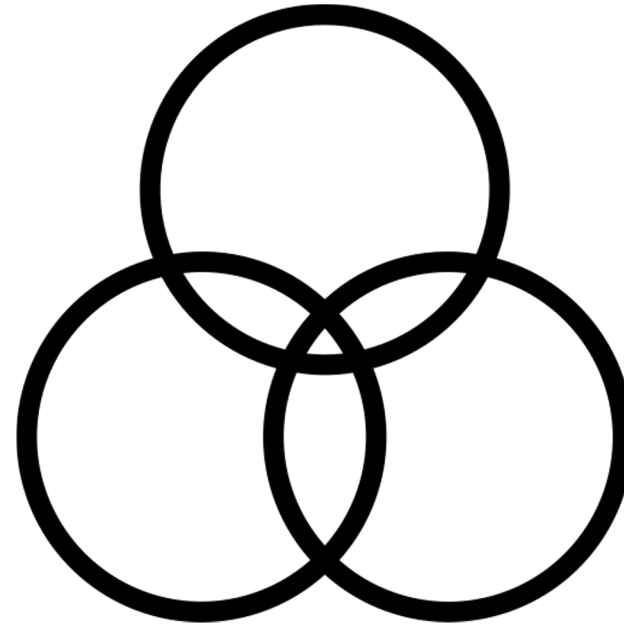
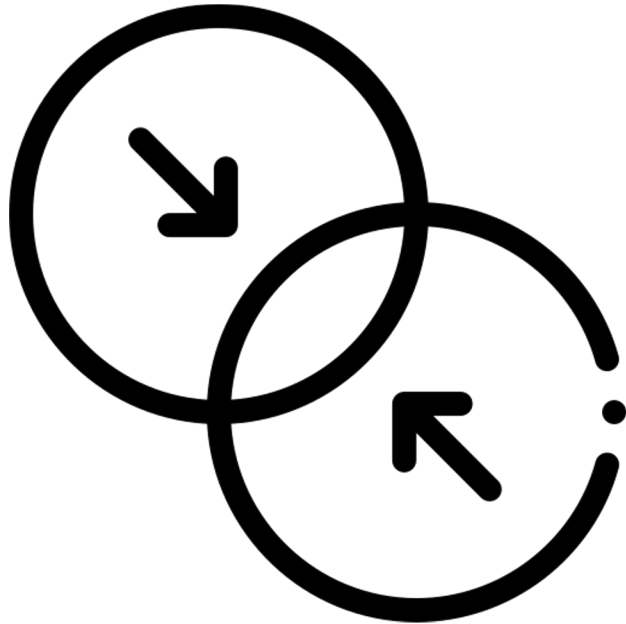
Hoang-Bach Ngo - TA
Minh-Loi Nguyen - STA

Objectives

Understand the concept of ensembling

Explore different ensembling methods

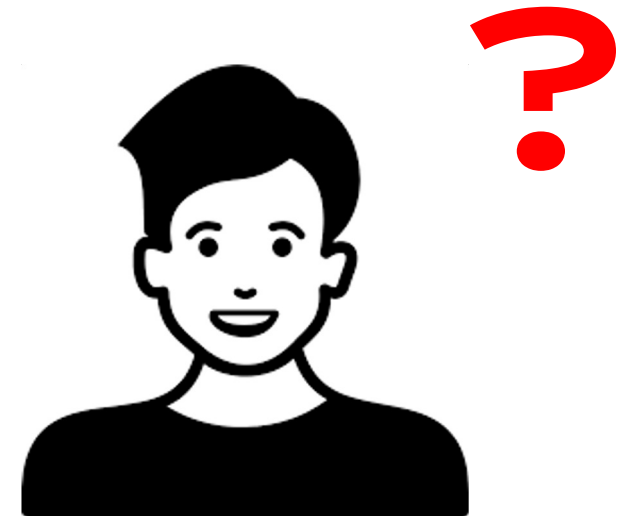
Learn the advantages of using ensembling techniques in machine learning



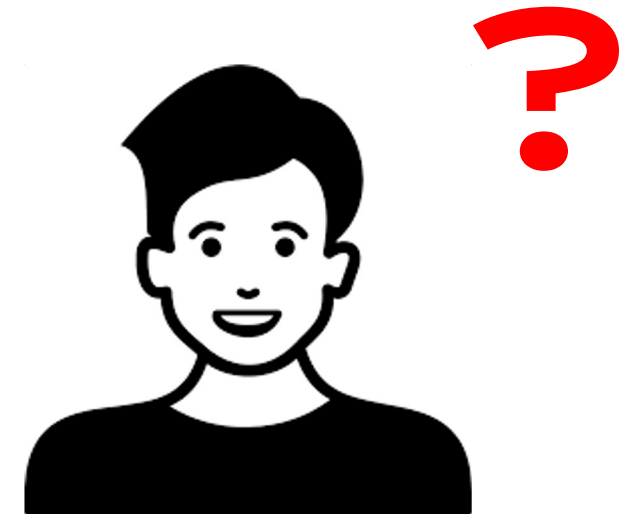
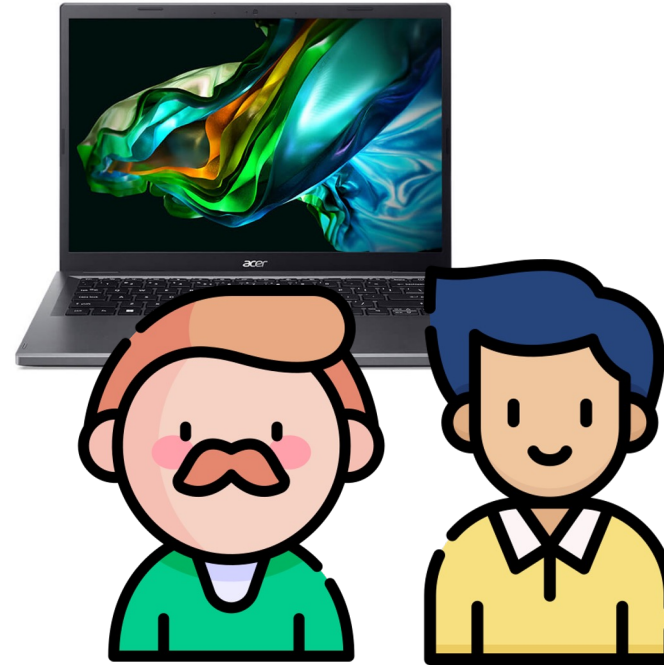
Why?



Scenario: Buying laptop



Why?



Scenario: Buying laptop

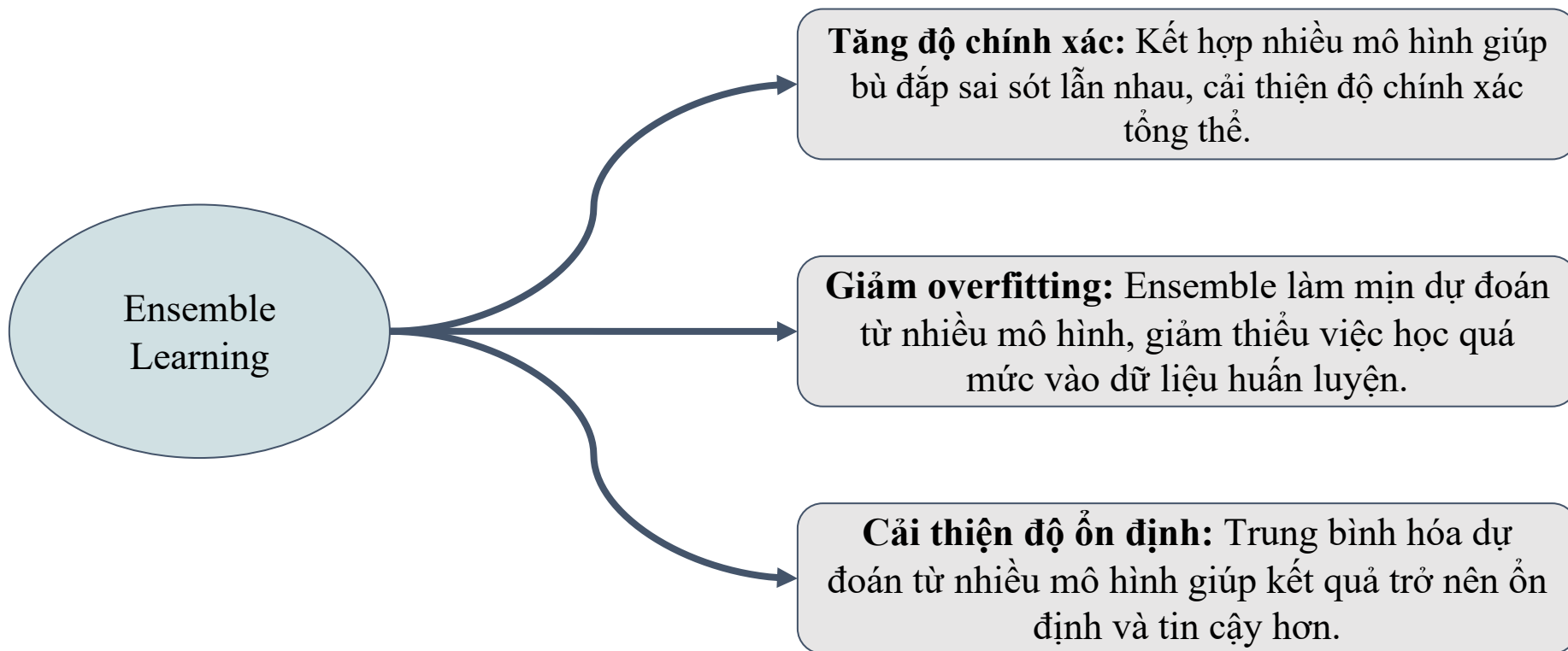
Why?



Scenario: real life voting



Why?



Why?

AI City Challenge 2024

Model Ensembling We employ the WBF method [34] to **ensemble** multiple models, using an IOU threshold of 0.75 and a skip bounding box threshold of 0.15. As illus-

CPA	Ensemble	Score correction	mAP
			0.4049 (baseline)
✓			0.4383 (+3.34%)
✓	✓		0.4591 (+2.08%)
✓	✓	✓	0.4792 (+2.01%)

3.3. **Ensemble** Module

3.3.1 Weighted Box Fusion

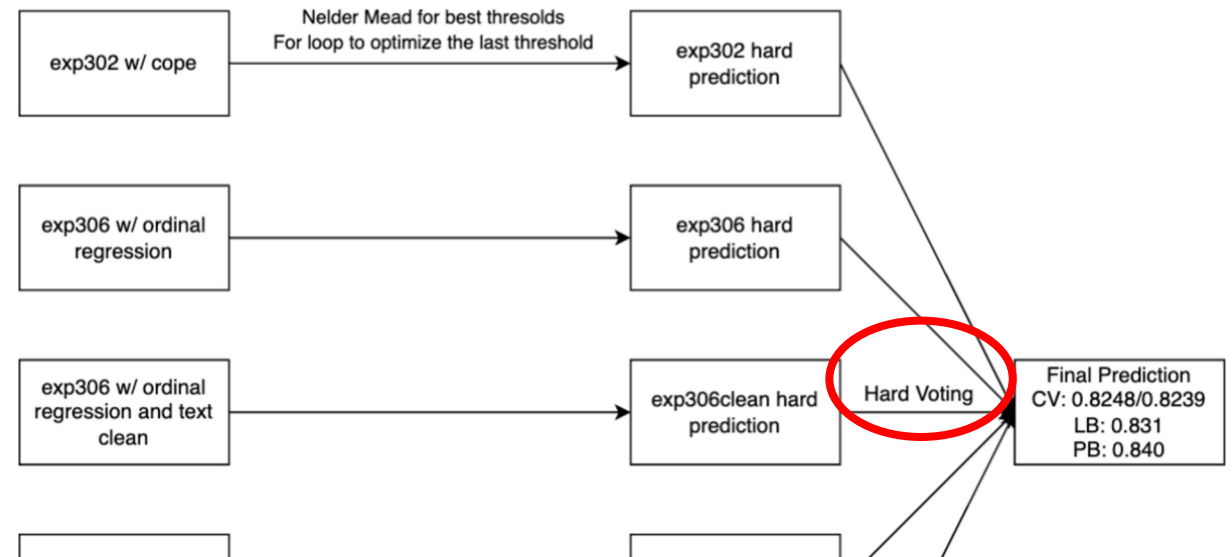
In tackling the multifaceted challenges inherent in helmet violation detection within diverse and complex environments, our **ensemble** module is pivotal in refining detection accuracy while effectively mitigating false negatives

Why?

Kaggle Competitions

Ensemble

From trained models I created ensembles via simple averaging, thresholding and fitted ensemble weights with just 4.5k samples weights independently even small ensembles fit the data way too



Final Submission

My final submission was a large **ensemble** (average of 45 predictions):

- 3 models: `microsoft/deberta-large`, `microsoft/deberta-v3-large`, `Qwen/Qwen2-1.5B-Instruct`
- 5 folds

Outline

1

**Voting
ensembles**

2

Stacking

3

**Ensemble in other
tasks**

4

Examples

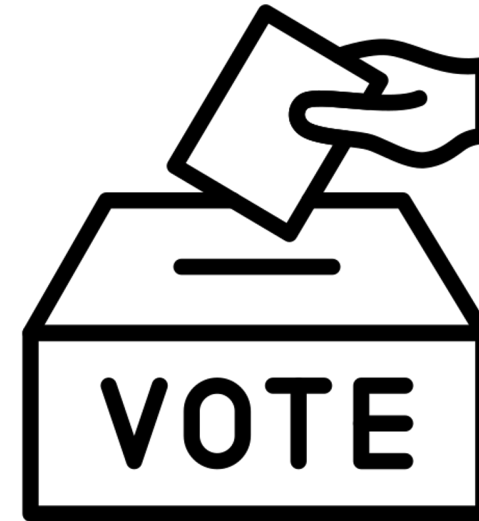
Outline

1 **Voting ensemble**

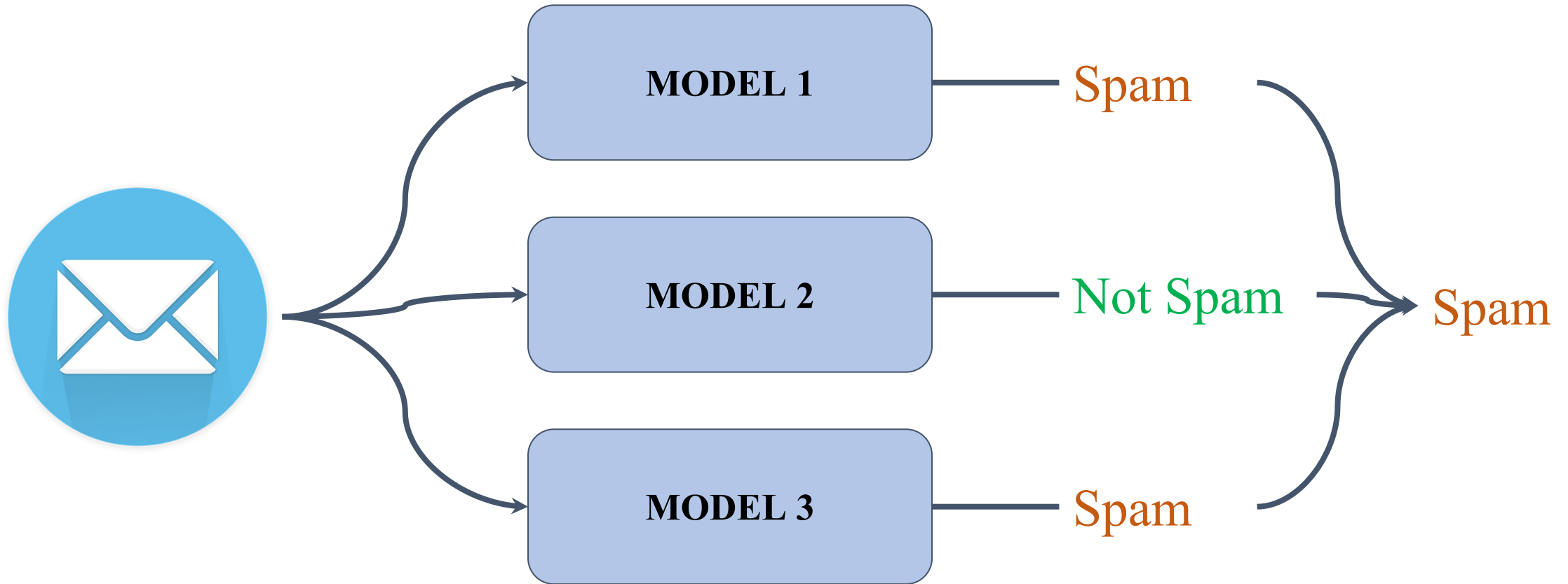
2 **Stacking**

3 **Ensemble in other tasks**

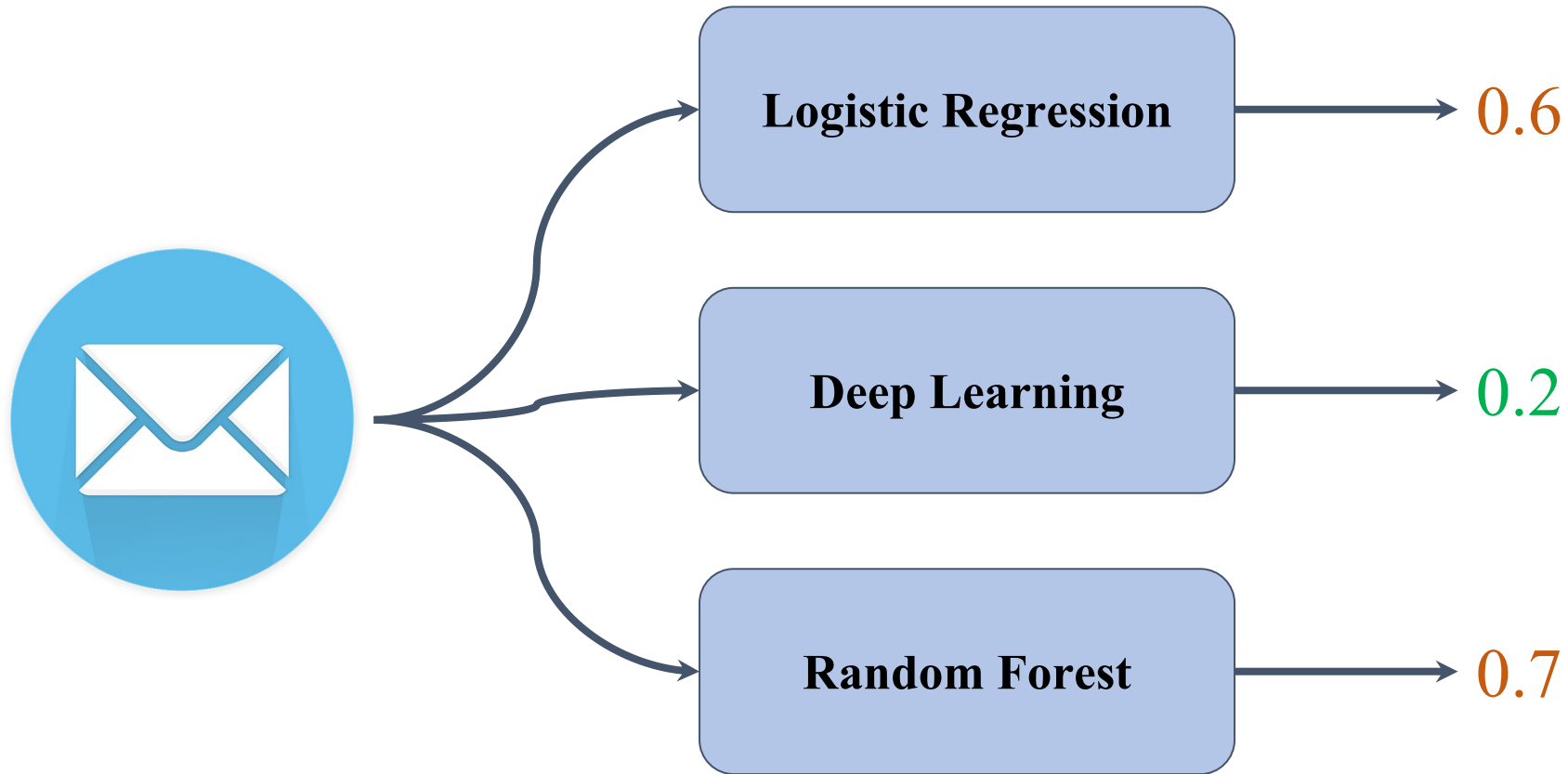
4 **Examples**



Voting ensembles

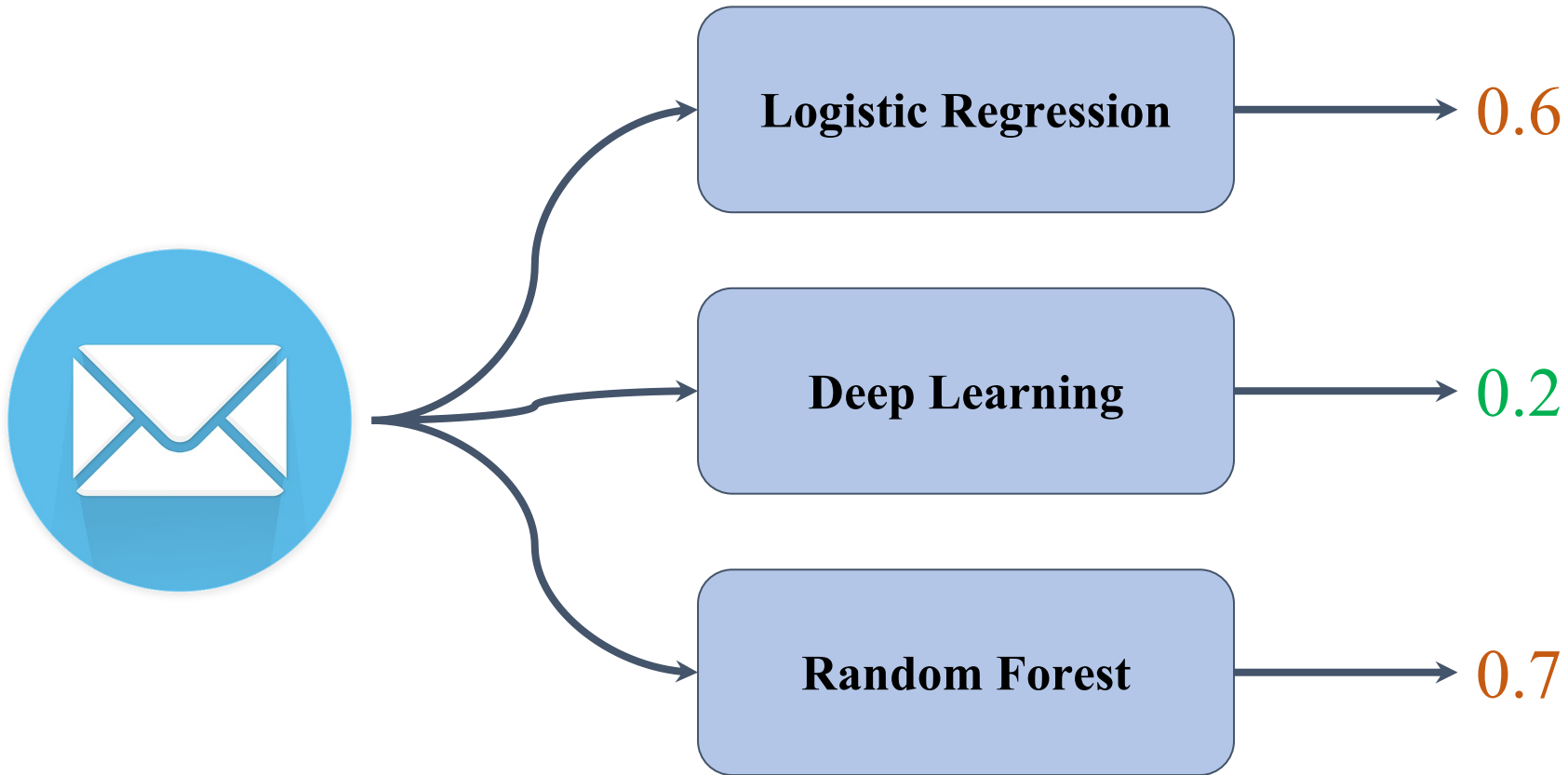


Voting ensembles



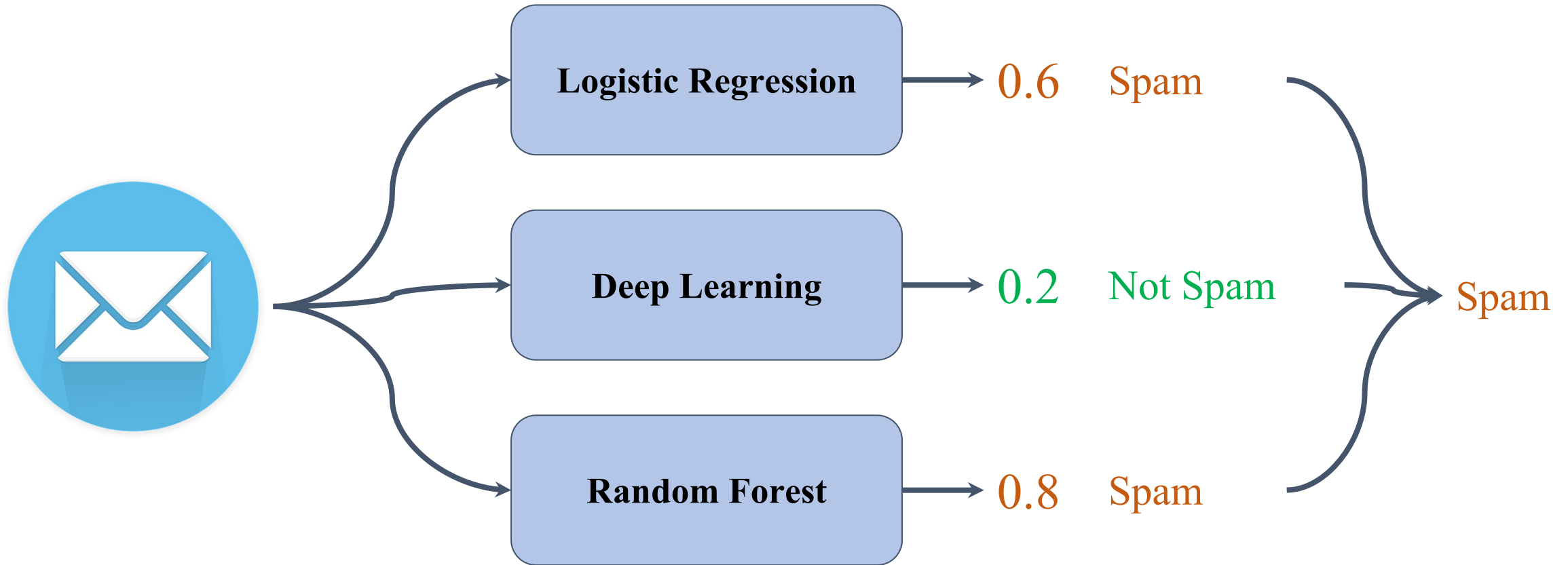
Voting ensembles

❖ Hard voting



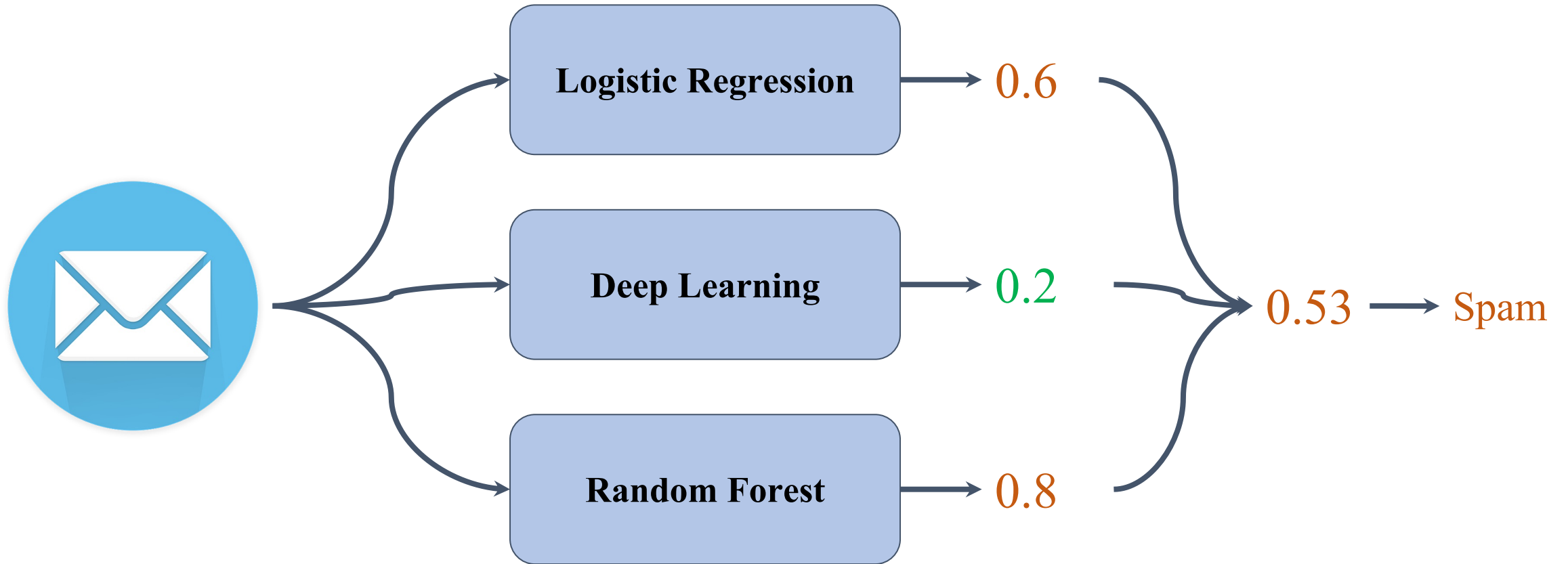
Voting ensembles

❖ Hard voting



Voting ensembles

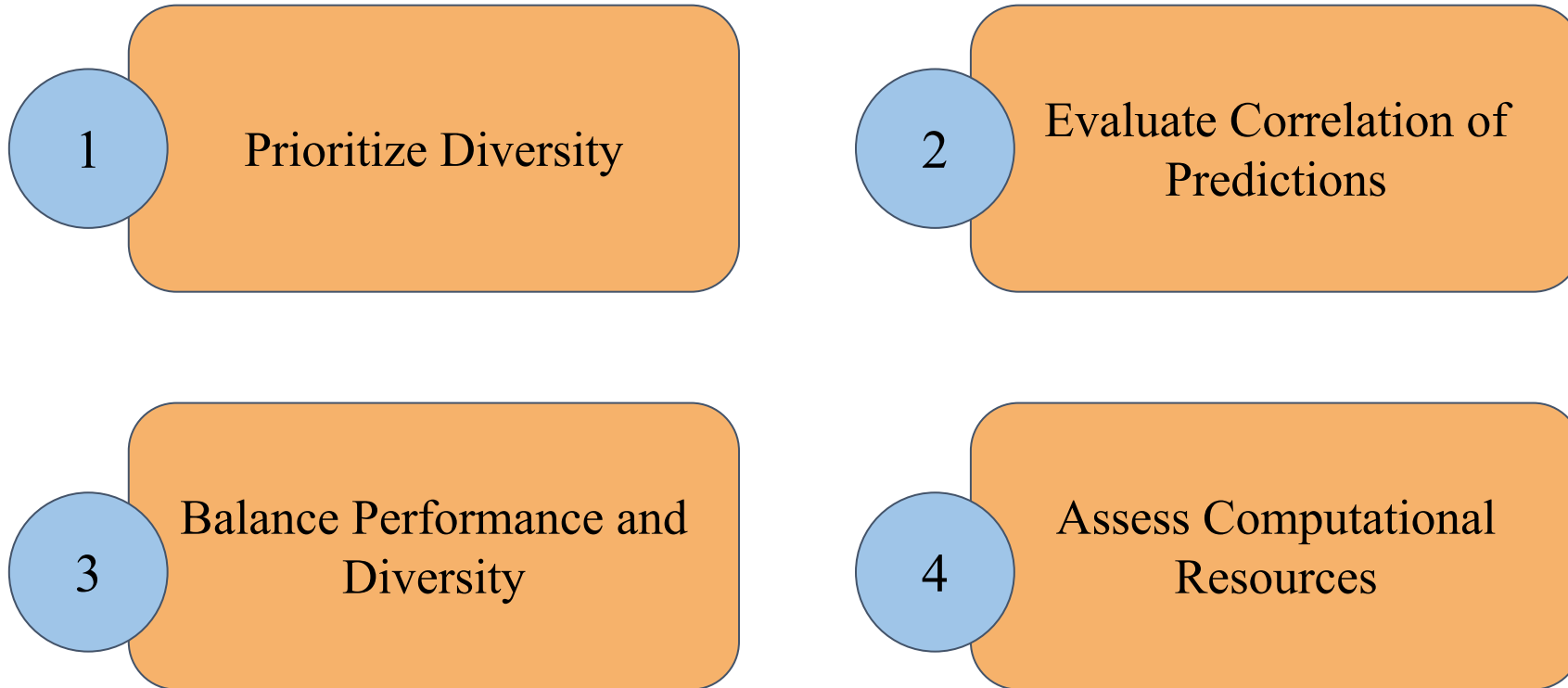
❖ Soft voting



Voting ensembles

	Hard Voting	Soft Voting
Input	Class predictions	Probability
Confidence Measure	No	Yes
Tie-Breaking	Needs explicit rule	Natural (via probabilities)
Sensitivity	High to classification threshold	Lower to classification threshold
Interpretability	More intuitive	Less intuitive
Task	Classification only	Classification and regression
Best Use Case	Equally reliable models	Varying model performance

Tips & Tricks



Outline

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Voting ensemble

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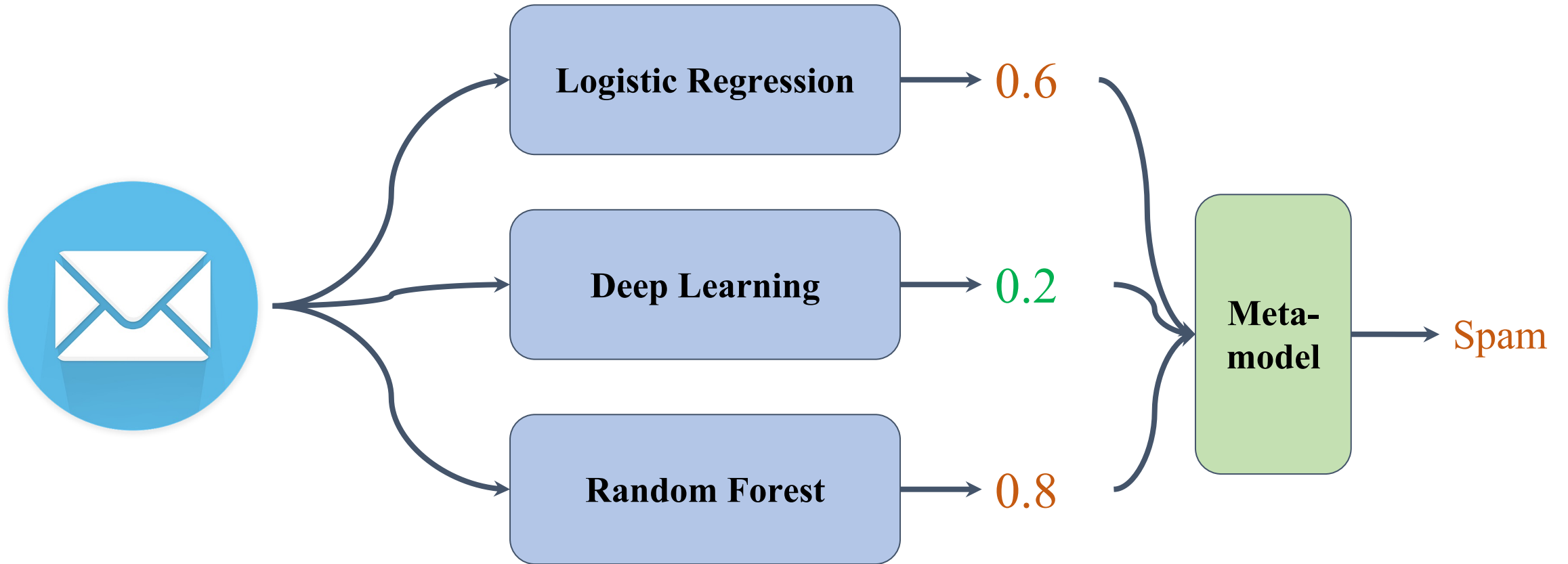
Ensemble in other tasks

4

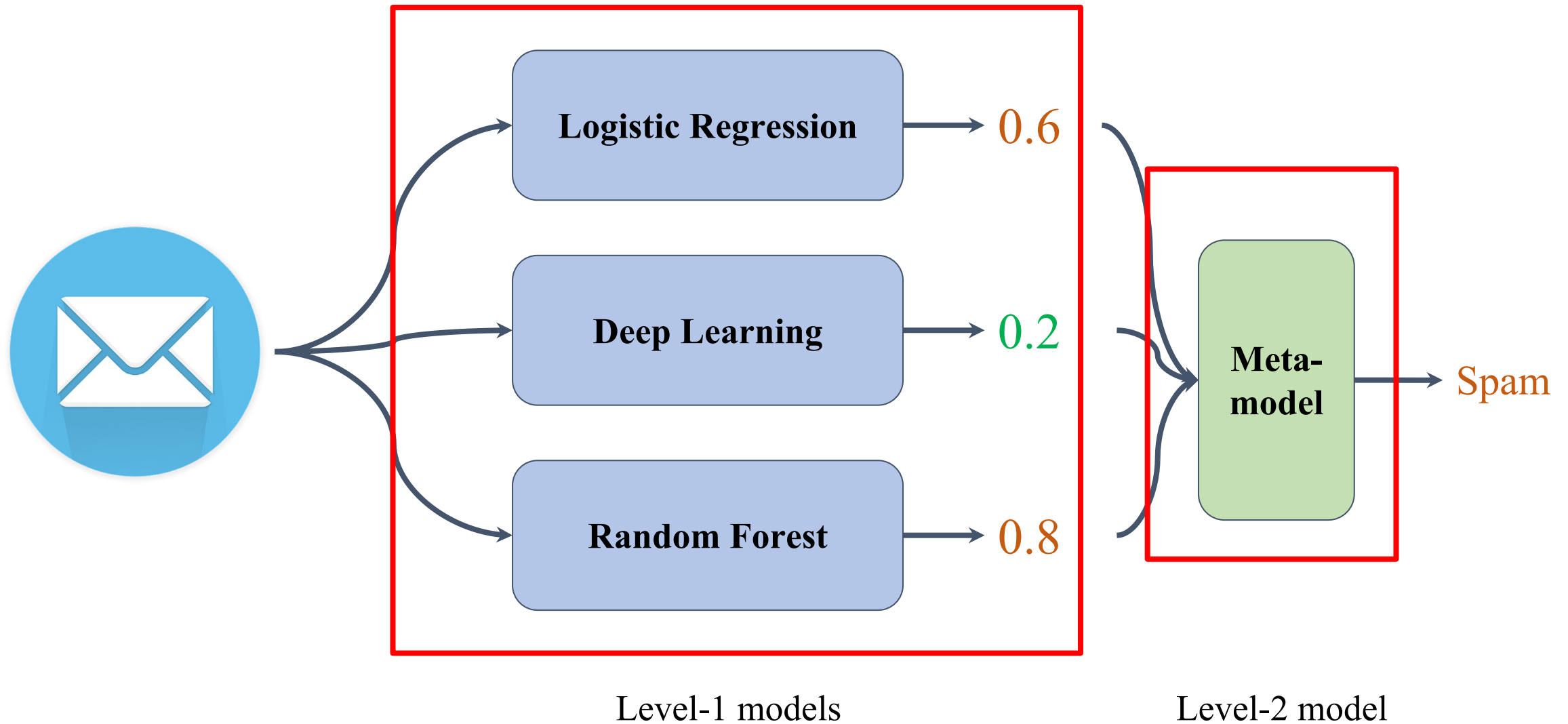
Examples



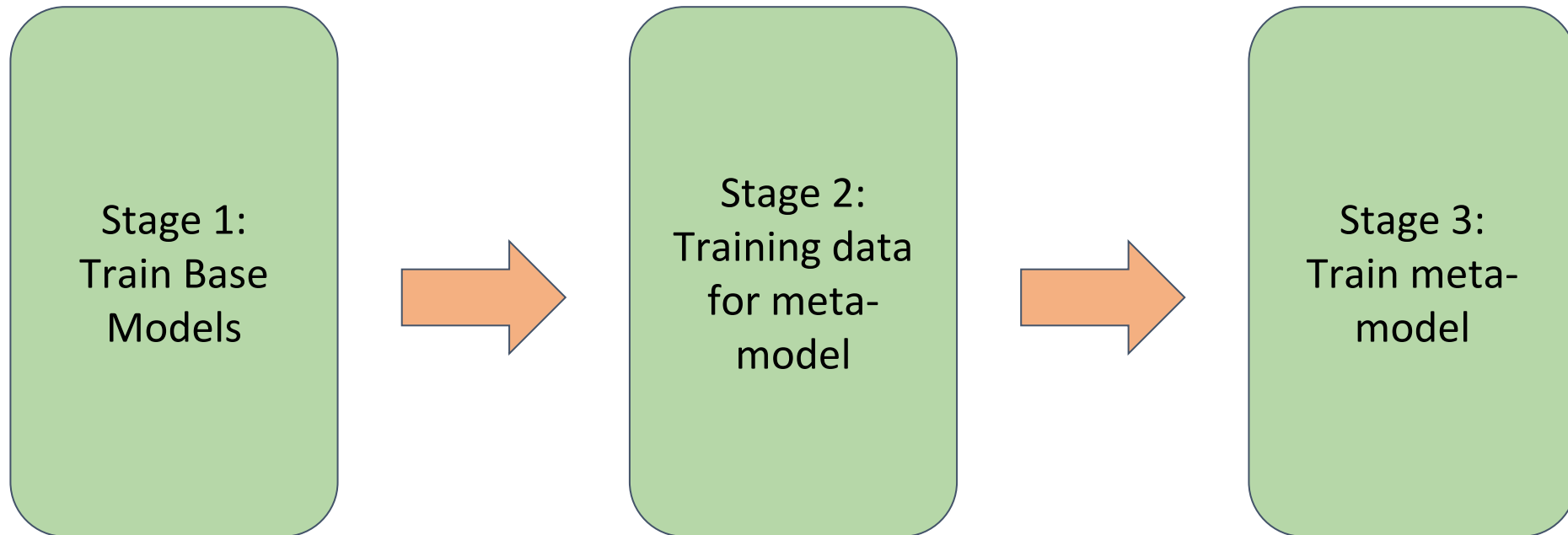
Stacking ensembles



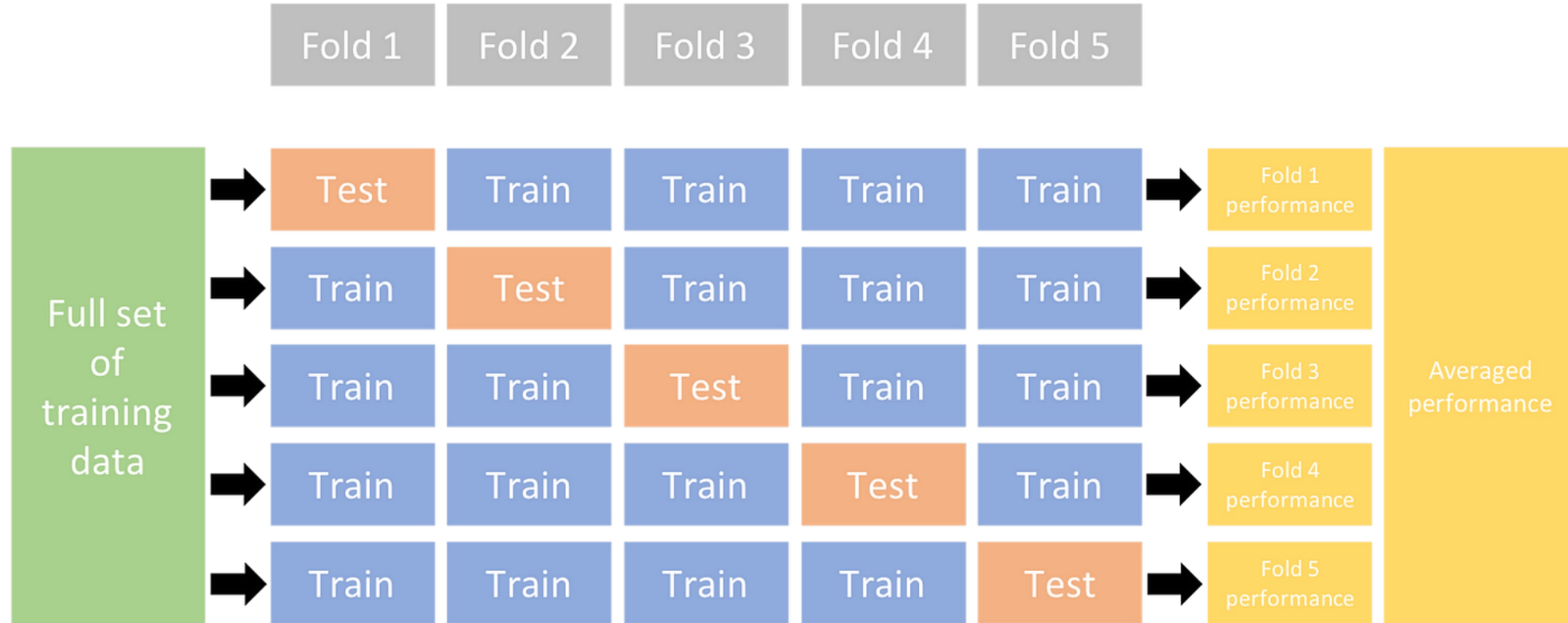
Stacking ensembles



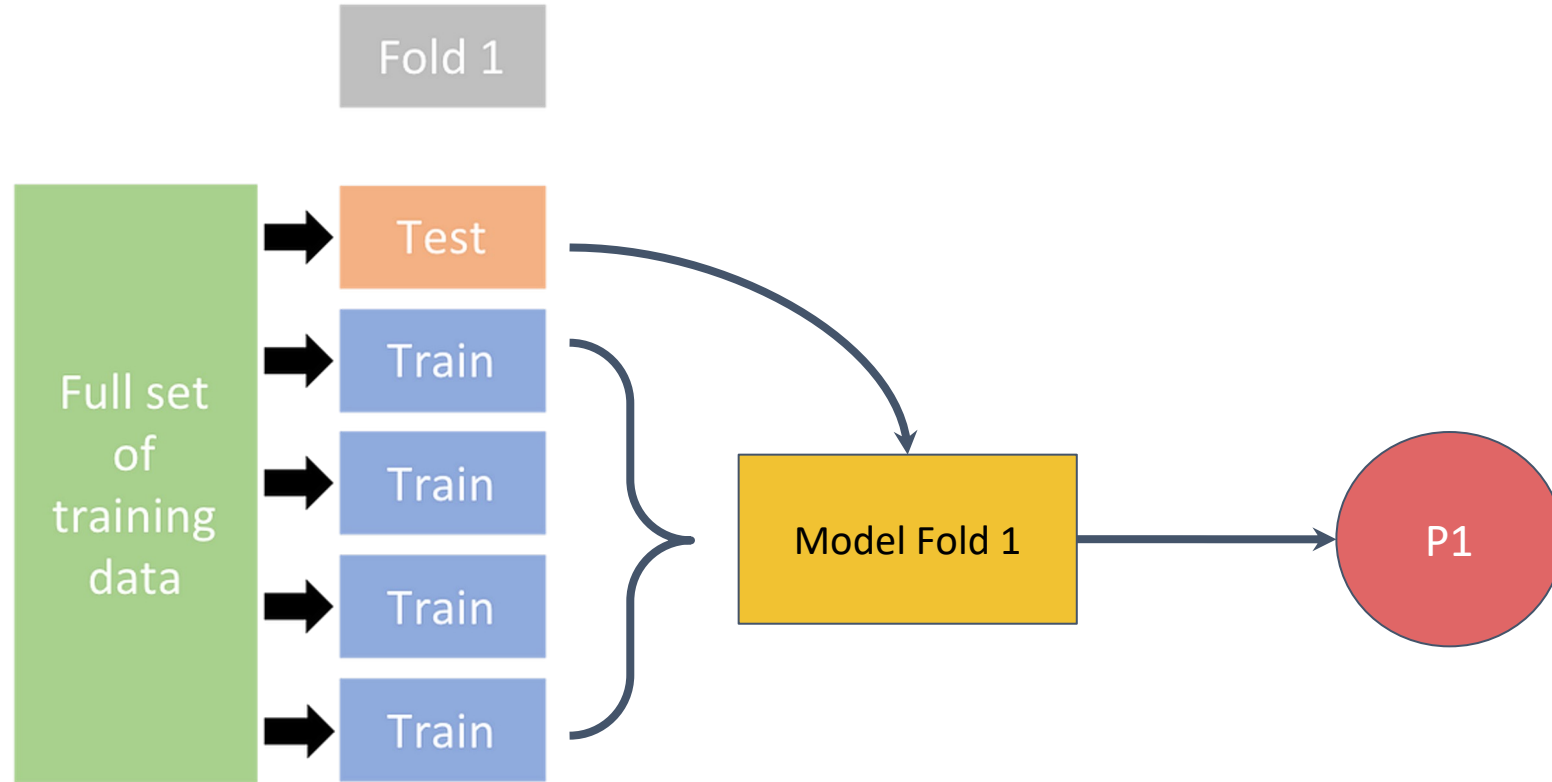
Stacking ensembles



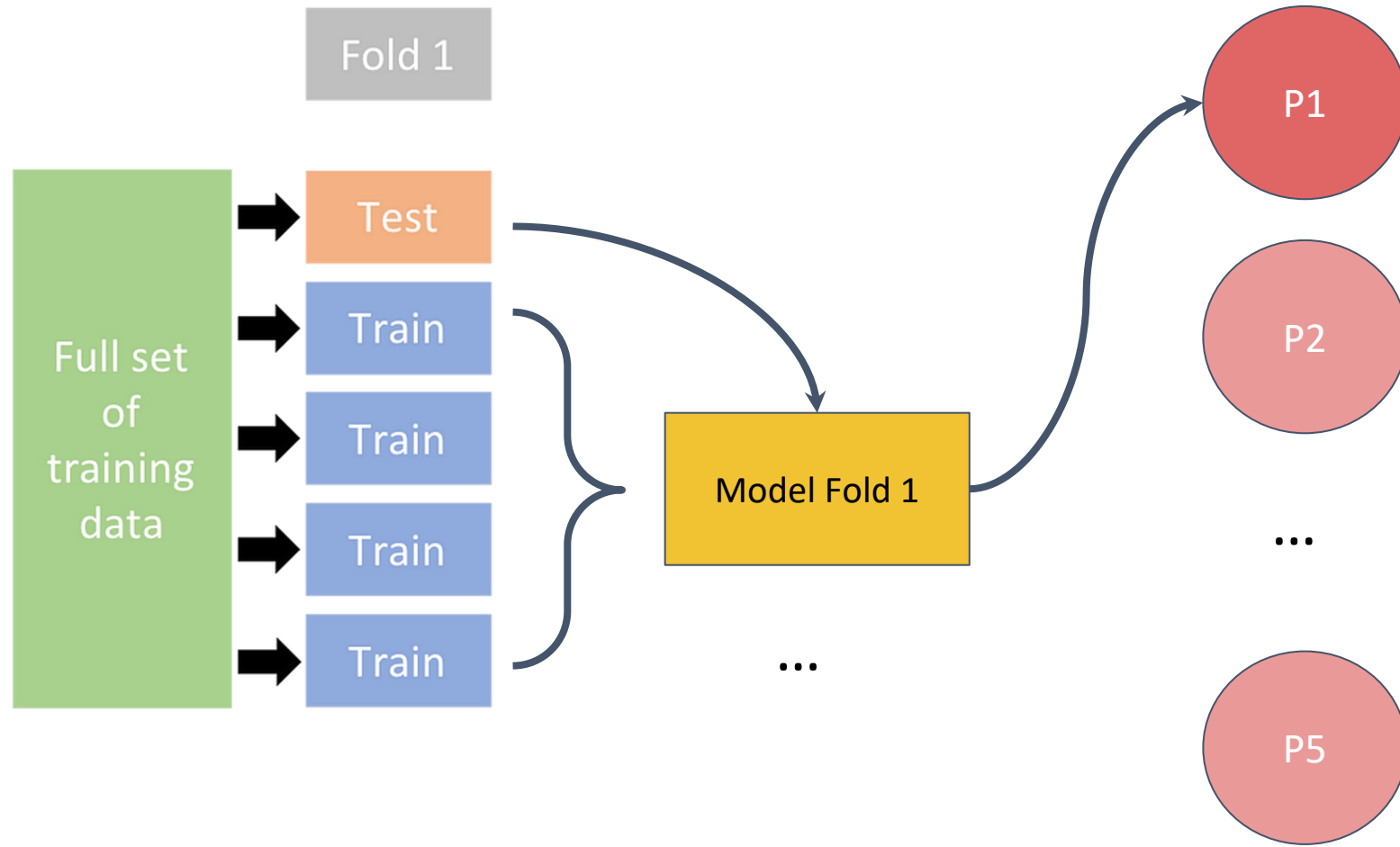
Stacking ensembles + KFold



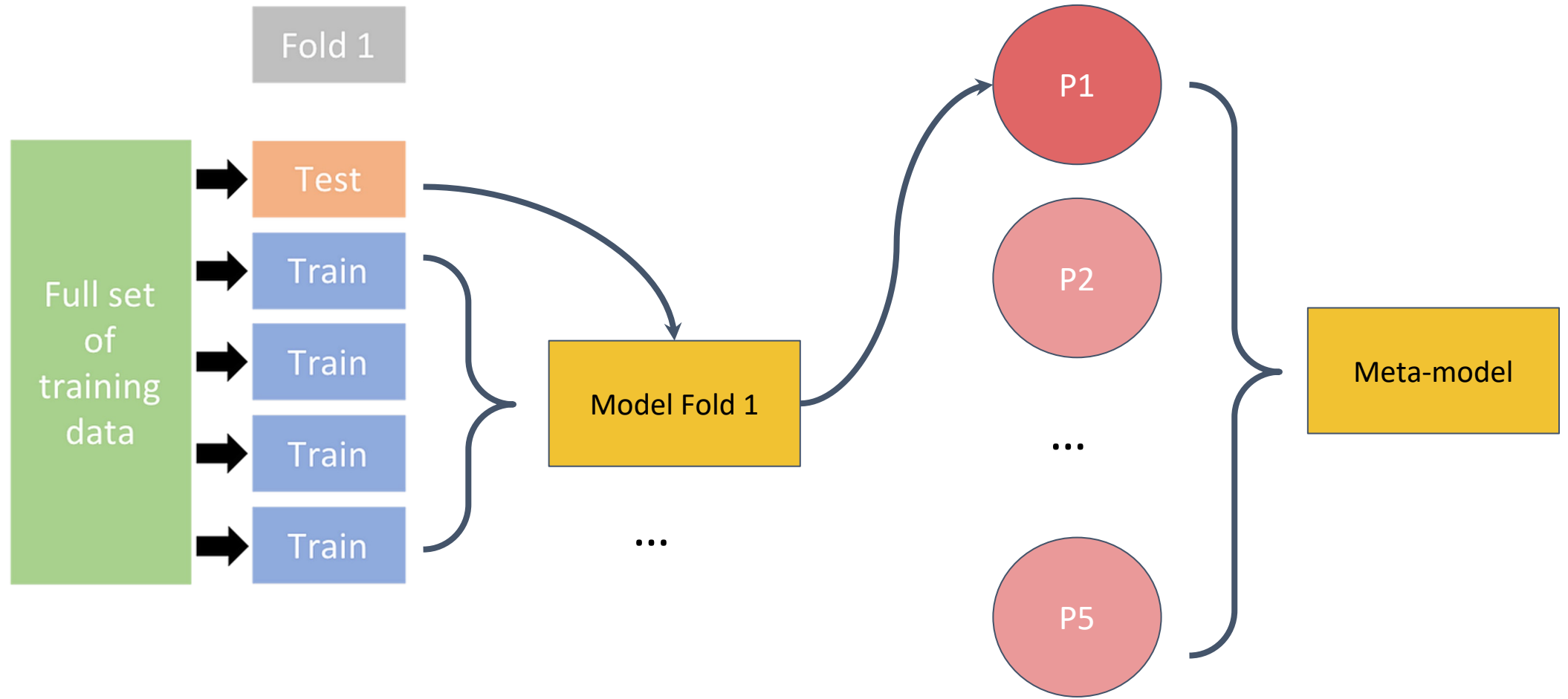
Stacking ensembles + KFold



Stacking ensembles + KFold

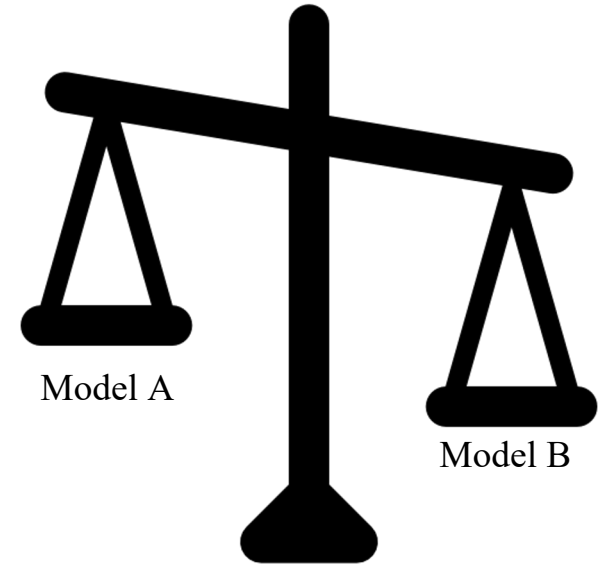


Stacking ensembles + KFold

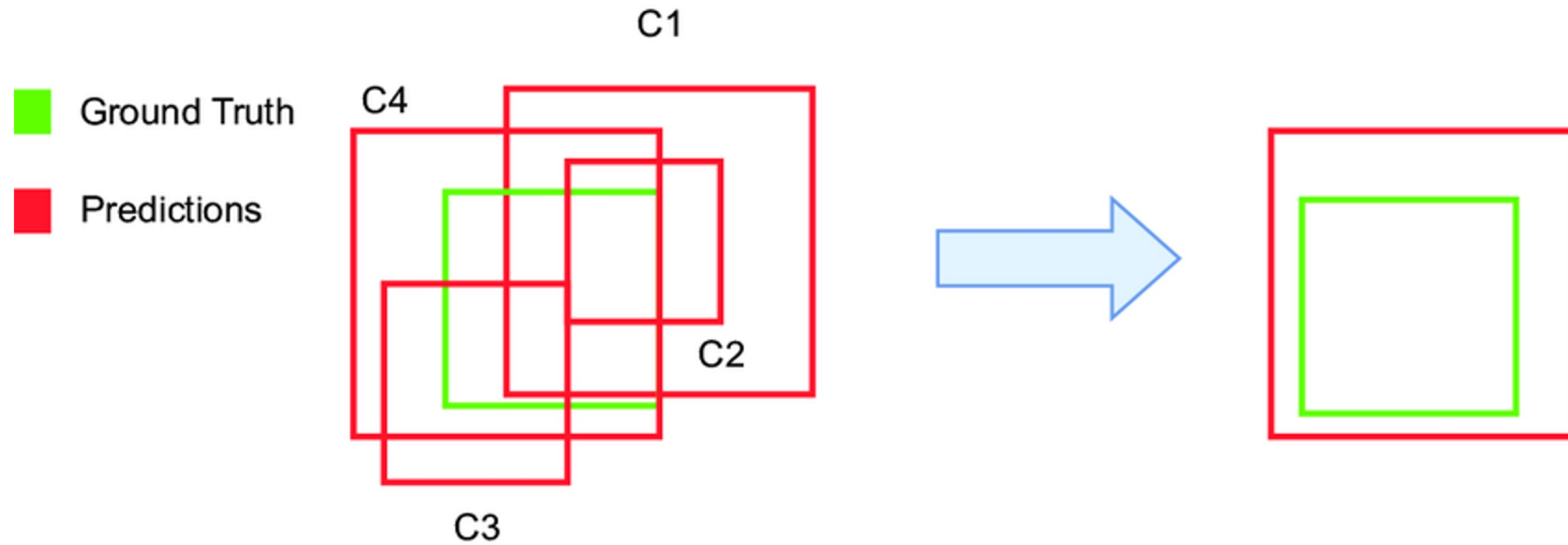


Outline

- 1 **Voting ensemble**
- 2 **Stacking**
- 3 **Ensemble in other tasks**
- 4 **Examples**



Ensemble in Object Detection



Boosting



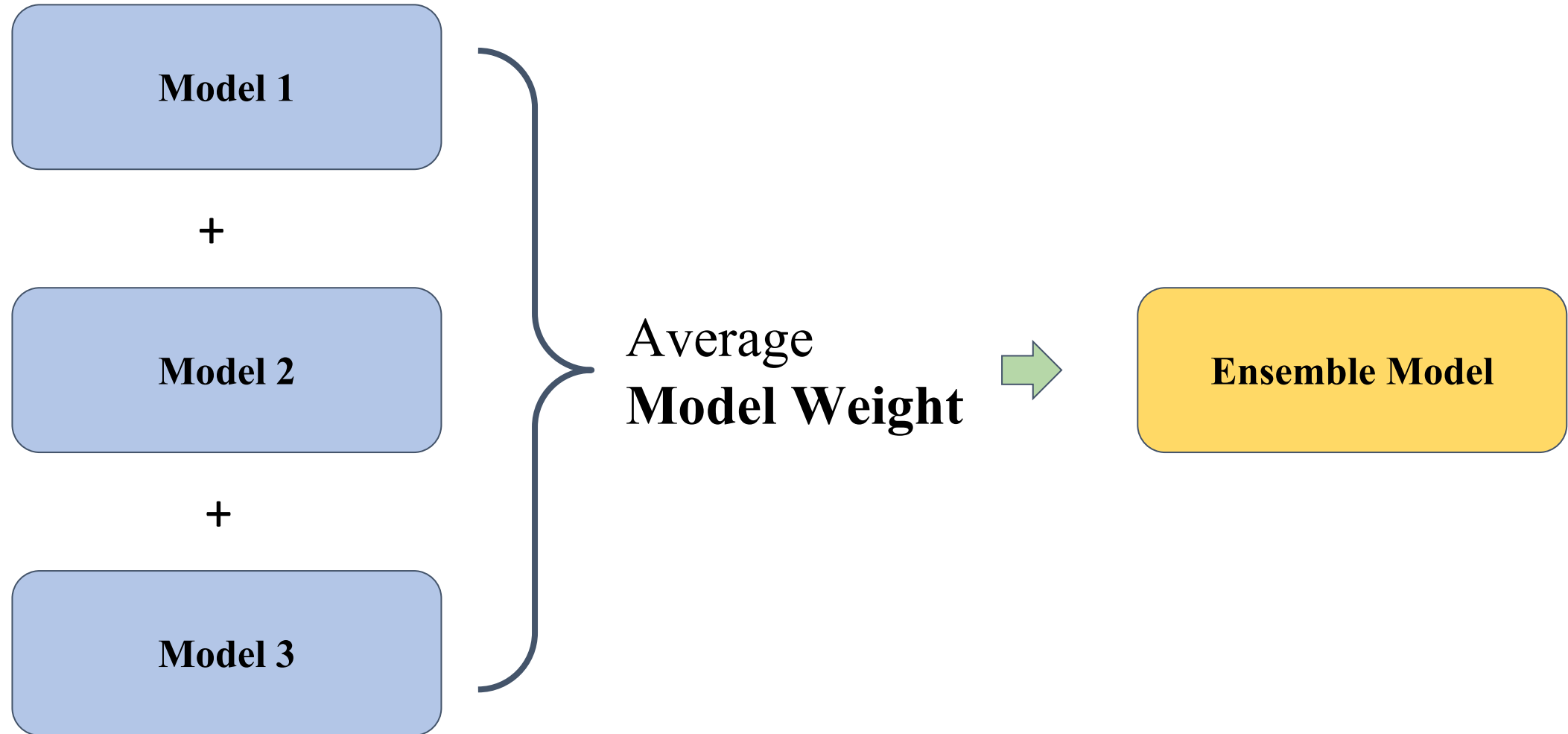
LightGBM

XGBoost



CatBoost

Model Soups



Model Soups

Pros	Cons
Improved performance over individual models	Limited to similar model architectures
Efficient inference (single forward pass)	Potential loss of model diversity
Combines benefits of multiple training runs	Less flexible than some other ensemble methods

Outline

1

Voting ensemble

2

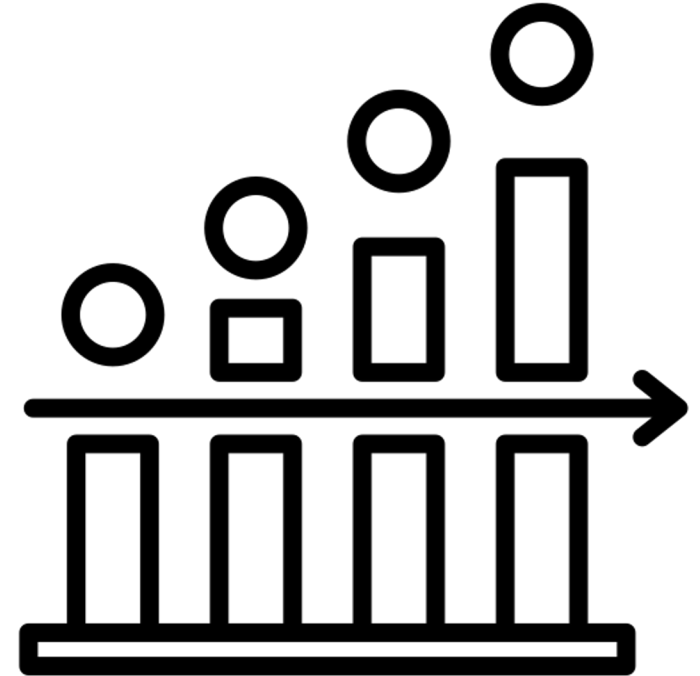
Stacking

3

Why Ensemble Works?

4

Examples





 KAGGLE · PLAYGROUND PREDICTION COMPETITION · 7 DAYS TO GO

Submit Prediction

...

Binary Classification of Insurance Cross Selling

Playground Series - Season 4, Episode 7

Overview

Data

Code

Models

Discussion

Leaderboard

Rules

Team

Submissions



Overview

Welcome to the 2024 Kaggle Playground Series! We plan to continue in the spirit of previous playgrounds, providing interesting and approachable datasets for our community to practice their machine learning skills, and anticipate a competition each month.

Your Goal: The objective of this competition is to predict which customers respond positively to an automobile insurance offer.

Start

24 days ago

Close

7 days to go

Evaluation

Submissions are evaluated using area under the ROC curve using the predicted probabilities and the ground truth targets.

Submission File

Competition Host

Kaggle

Prizes & Awards

Swag
Does not award Points or Medals

Participation

5,640 Entrants
1,867 Participants
1,745 Teams
11,021 Submissions

Tags

Beginner

Tabular

Binary Classification

Roc Auc Score

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LIVE DEMO

QUIZ TIME

